

Yuni Wu

South San Francisco, CA

yunihsuanwu@gmail.com — (720) 487-5403 — linkedin/wih — github/yuni-wyx — portfolio

Summary

Data engineer focused on ETL/ELT, data quality monitoring, validation workflows, metadata pipelines, BigQuery analytics, Cloud Run/Pub/Sub workflows, relational data modeling, and query optimization.

Technical Skills

Languages: Python (advanced), SQL, Go, R

Data Engineering: ETL/ELT, schema validation, relational data modeling, batch/streaming ingestion, metadata pipelines

Data Quality: anomaly detection, regression monitoring, validation workflows, observability, structured logging

Cloud & Storage: GCP (BigQuery, Cloud Run, Pub/Sub, Cloud Storage), Docker, PostgreSQL, MongoDB

Systems: distributed systems, asynchronous processing, event-driven workflows, query optimization, partitioning, clustering

Projects

NailAI — Serverless AI Microservice Platform

[demo]

- Built event-driven AI processing workflows using Cloud Run, Pub/Sub, and Cloud Storage for asynchronous image inference.
- Achieved 0% failure at 200 concurrent users with idempotent orchestration, retry handling, and Pub/Sub job routing.
- Designed request, job, and prediction metadata pipelines for querying, monitoring, and downstream analytics.
- Optimized BigQuery partitioning and clustering for warehouse-centric downstream analysis and lower query latency.

FitMatch — Product Size Recommendation Platform

- Developed a FastAPI/Next.js recommendation platform with PostgreSQL schemas, catalog ingestion workflows, product-profile mappings, and internal QA surfaces.
- Imported and validated Tienhuang catalog data across 1,205 apparel products, 65 size profiles, 407 profile chart rows, and 1,205 product-profile relationships.
- Created validation workflows, feedback collection, and regression-style boundary tests, achieving 24/24 passing validation cases.

Stock Intelligence Copilot — Agentic Stock Research Platform

[demo]

- Implemented FastAPI-based streaming workflows and structured execution schemas that improved traceability, debugging, and observability across analytical tasks.
- Integrated market data, fundamentals, news, and earnings tools into evidence bundles with per-tool error capture and fallback-safe API presentation.

Experience

Software Engineer

June 2021 – Aug 2022

Shopee Co. Ltd.

- Engineered SQL-based validation and anomaly detection pipelines across 800+ production scenarios, reducing manual QA workload by 80%.
- Built data quality monitoring pipelines to detect regression anomalies and surface near real-time reporting inconsistencies, accelerating issue detection by 78%.
- Optimized distributed ETL workflows and GCP data access patterns, reducing query latency by 15%.
- Developed automated data collection workflows using APIs and Selenium-based browser automation to support validation, monitoring, and large-scale data quality analysis.

Machine Learning Engineer Intern

May 2025 – Aug 2025

MyEdMaster LLC

- Analyzed Cloud Run inference architectures with Docker and Cloud Storage-backed asynchronous processing to assess deployment tradeoffs.
- Implemented monitoring, regression validation, and structured logging workflows for Cloud Run inference evaluation.

Graduate Research Assistant

Sept 2025 – May 2026

CAIRO Lab, University of Colorado Boulder

- Developed robotics data processing pipelines to convert rosbag recordings into structured datasets for embodied AI navigation experiments.
- Processed ROS telemetry and perception outputs into state-action datasets for navigation analysis, reliability evaluation, and inference latency benchmarking.

Education

University of Colorado Boulder, MS, Data Science GPA: 3.76/4.0

May 2026

Soochow University, BS, Data Science GPA: 3.7/4.0

June 2021